Global Environmental Concerns

Population Growth

Robert McNamara(1973), wrote *One Hundred Countries, Two Billion People*

Rapid population growth was a threat that would have "catastrophic consequences"

James Wolfensohn,
World Bank president from 1995 to 2005

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Robert Zoellick delivered a speech identifying six strategic themes to meet global challenges. Reducing population was not one of them; in fact, "population growth" is not mentioned even once

Robert McNamara, *One Hundred Countries, Two Billion People* (New York: Praeger, 1973), p. 31.

Global Environmental Concerns

Population Growth

Last Ice Age, about 13,000b years ago, when humans on all continents were still living as hunter–gatherers, or 12,000b years ago with the first signs of agricultural settlements,
or 7,000m years ago with the first indications of urbanization.

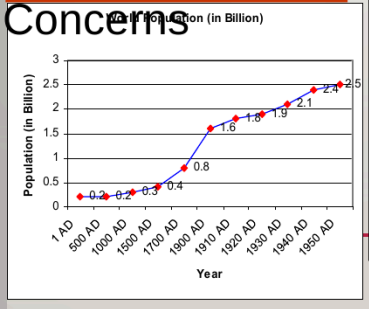
From the beginning of human settlements, it took more than 10,000 years for the world's population to reach 1 billion in 1820
1 billion people every 12 to 15 years. No wonder McNamara was so worried.

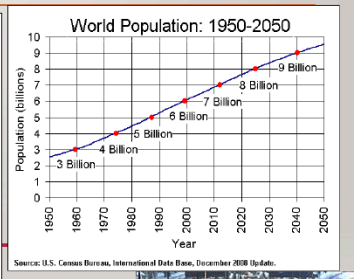
During the Industrial Revolution (late eighteenth and early nineteenth centuries), the
food supply grew and became more reliable. The death rate fell, life expectancy increased, and population growth gradually accelerated.

The Industrial Revolution, which marked the start of modern economic growth, further expanded the Earth's **population-carrying capacity**.

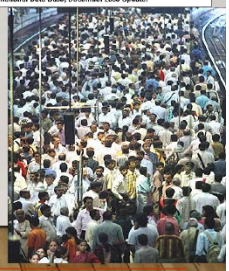
Global Environmental Concerns

Population Growth



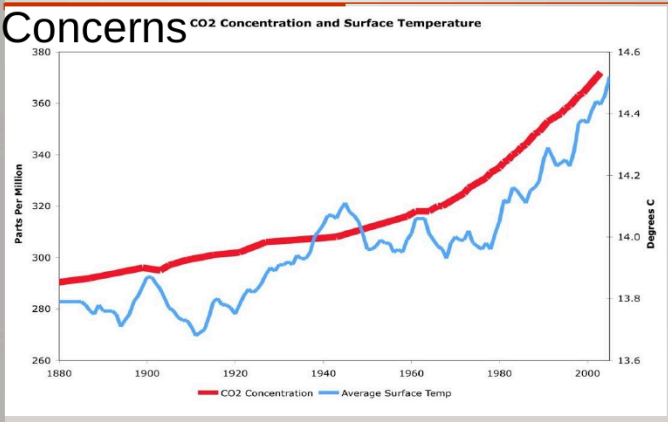


- four or five children are born in every second
- two other people die in same time
- nearly 2.5 more humans per second added to the world population.
- This means, we are growing almost 83 million more people per year.



Populatio n Growth

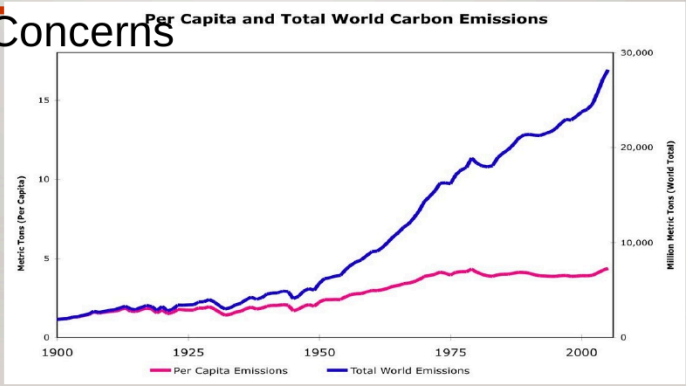
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This chart shows the correlation between Atmospheric Carbon Concentration (in Parts Per Million) and Average Surface Temperature.

Populatio n Growth

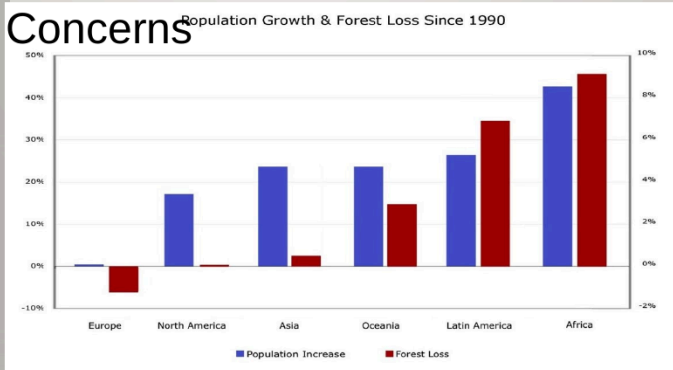
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A curious case of divergence. This chart shows how Total Worldwide Carbon Emissions have increased at an Exponential Rate, while Per Capita Carbon Emissions have grown at a Linear Rate during the past century. To what can we attribute the difference in the rate of growth?

Populatio n Growth

Global Environmental Concerns



This very important chart shows the relationship between Population Growth and deforestation. Conclusion: The faster the population grows, the faster the forests disappear.

Populatio n Growth

Global Environmental



Populatio n Growth

Global Environmental Concerns

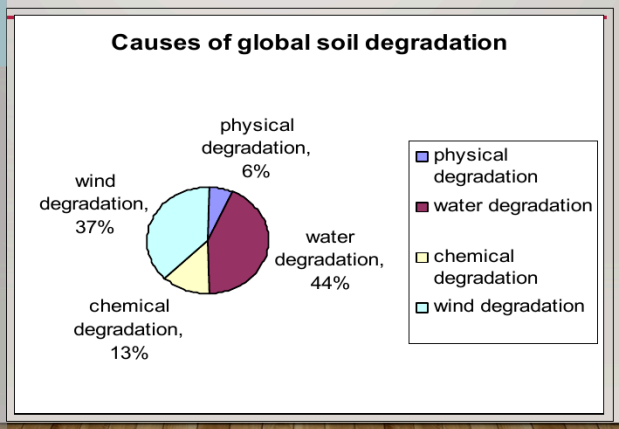
Impacts of Population Growth

- Great pressure is being placed on arable land, water, energy, and biological resources to provide an adequate supply of food while maintaining the integrity of our ecosystem.
- Overload the Earth's life-support systems
- Vital resources are stressed by the **dual demands** of increasing population and increasing consumption per person
-

- Groundwater supplies are being depleted, agricultural soils are being degraded, oceans are being overfished, oil reserves are being drawn down, and forests are being cut faster than they can regrow
- Human population inevitably outstrips their food supply and eventually collapses into starvation, crime, and misery
- Sometimes it is viewed that the root cause of environmental degradation is **inequitable distribution of wealth and power rather than population size**

Global Environmental Concerns

Soil Degradation



Soil Degradation Processes, Causes, and Assessment Approaches, Table 1 The global extent of human-induced soil degradation. (Modified from Oldeman et al. 1991)

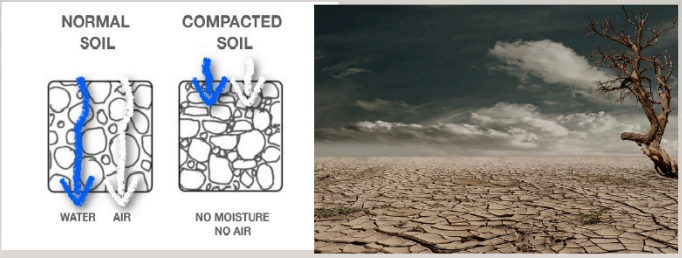
Region	Land surface (10^6 ha)		Land surface affected by different types of soil degradation (10^6 ha)			
	Total	Degraded	Water erosion	Wind erosion	Chemical soil degradation	Physical soil degradation
Africa	2,966	494	227	186	62	19
Asia	4,256	748	441	222	74	12
South America	1,768	243	123	42	70	8
Central America	306	63	46	5	7	5
North America	1,885	96	60	35	–	1
Europe	950	218	114	42	26	36
Oceania	882	102	83	16	1	2
World	13,013	1965	1,094	548	240	83

"–" indicates that an eligible area is affected by the specific degradation process

Global Environmental Concerns

Soil Degradation

Physical degradation includes compaction by heavy machinery or trampling by cattle, water accumulation from excess irrigation and poor drainage, and laterization (solidification of iron- and aluminum-rich tropical soil when exposed to sun and rain).



SOIL DEGRADATION

- Deforestation: The main natural causes are fires and floods, and among human activities, the causes are logging, timber production, the conversion of a forest into agricultural land, and urbanization.
- Shifting cultivation: It is an old farming practice where the “slash-and-burn” technique is applied to clear land, followed by a long fallow period, which is important for the restoration of soil fertility
- Soil erosion is the main and most widespread form of land degradation. Soil erosion is caused by the activity of water and wind and represents the detachment and movement of soil particles from one place to another. This process can be natural or accelerated by human activity. It depends on many factors, among which the most important are the configuration of the terrain (slopes) and climatic-meteorological conditions.



- Salinization is one of the most widespread forms of soil degradation and occurs in arid and semi arid areas, where the amount of precipitation is small and irrigation is applied without a proper drainage system. However, salinization can occur in all climatic areas if irrigation is irregularly applied.
- Acidification, like salinization, is a severe form of degradation that results in a reduction in the agricultural land production potential. The acidification process prevents plants using water, resulting in drying and erosion, and causing increased leaching of nutrients and an irreversible breakdown of silicate minerals in the soil(Fitzpatrick 2002).

- **Interstellar dust** in the atmosphere primarily affects Earth's climate by scattering and absorbing **shortwave and longwave radiation,** directly influencing surface temperature. Beyond that, it affects life on Earth through direct and indirect interactions with the biotic and abiotic components of the Earth system. Dust affects the very basic process of rainfall through its direct and indirect effects; thus, it regulates the hydrological cycle and the monsoon systems, which ultimately affect the distribution of water resources and agricultural production

TERRESTRIAL FACTORS

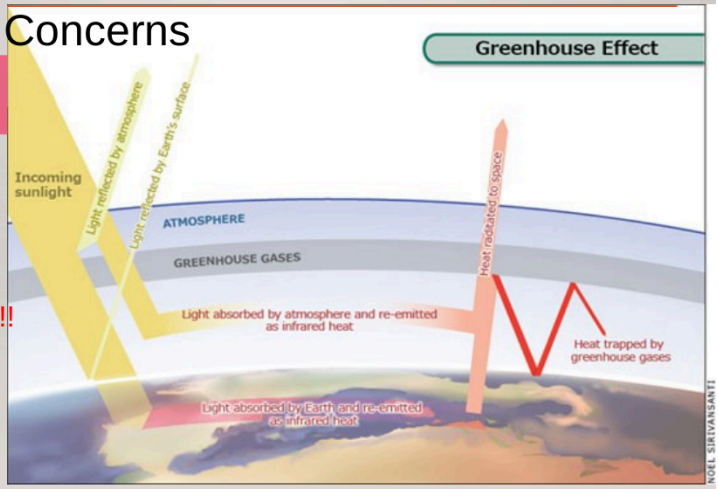
- Atmospheric chemistry plays a critical role in climate change by **controlling the amount and distribution of important climate forcing agents like water vapor, methane, and tropospheric and stratospheric ozone**.
- Atmospheric reflectivity: When sunlight reaches Earth, it can be reflected or absorbed. The amount that is reflected or absorbed depends on Earth's surface and atmosphere. Any change in surface albedo will alter climate by **drastically changing the amount of solar energy absorbed by the planet**. When the angle of the sun to the surface is low (closer to the horizon), solar energy is less intense since it is spread out over a larger area. The term **albedo** refers to the amount of solar radiation reflected from an object or surface, often expressed as a percentage. Earth has an albedo of about 30%, meaning that 70% of the sunlight that reaches the planet is absorbed. Absorbed sunlight warms Earth's land, water, and atmosphere.

- Surface Reflectivity: **The capability of a surface to reflect solar energy** is measured by its albedo or reflectivity in the visible portion of the energy spectrum. **Light-colored objects and surfaces, like snow and clouds, tend to reflect most sunlight,** while **darker objects and surfaces, like the ocean, forests, or soil, tend to absorb more sunlight.**

Global Environmental Concerns

Atmospheric Change

Our Concern!!!



Global Environmental Concerns

Impacts !!!

Atmospheric Change

Increased global temperature

At the current rate, global temperature may increase by 1.8–4.0ºC by the end of this century (IPCC 2007)

Greenhouse gases can influence animal populations through mechanisms

… can enter terrestrial and aquatic food webs and alter ecosystem functioning

Atmospheric contaminants commonly do this in two ways: 1) bottom-up and 2) top-down pathways

1

Alter plant anatomy and physiology

2

Alter population dynamics for their prey

Global Environmental Concerns

Biodiversity loss

Biodiversity loss

Defined as…

«*The long-term or permanent quantitative or qualitative reduction in components of biodiversity and their potential to provide goods and services, to be measured at global, regional and national levels*».

Loss of diversity (e.g., **Convention on Biological Diversity**, etc.),

Loss/diminished capacity for the components of diversity to provide a particular service (e.g., unsustainable harvest),

Homogenization of biodiversity (e.g., monoculture agriculture).

Example: Provisioning services



Medicines (Australian red-eyed tree frog)

Example: Cultural services



Hiking in the Alps, Switzerland

Example: Cultural services



Masai men, Kenya

Example: Regulating service



Lady bug and plant lice

Thanks

References

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Cunningham, William, Principles of Environmental, second edition

http//gadflyer.com

http://www.nature.com/scitable/knowledge/library/global-atmospheric-change-and-animal-populations